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Bepaal de nulpunten en de ontbinding in factoren in $\mathbb{R}[z]$ van:

$F(x) = z^8 + z^4 + 1$

Manier 1:

stel $t = z^4$

dan $F(x) = t^2 + t + 1$

$D = 1 - 4 = -3$

$$t_1 = \frac{-1-\sqrt{-3}}{2} = \frac{-1-i\sqrt{3}}{2} = \operatorname{cis}\left(\frac{-2\pi}{3}\right)$$
$$t_2 = \frac{-1+\sqrt{-3}}{2} = \frac{-1+i\sqrt{3}}{2} = \operatorname{cis}\left(\frac{2\pi}{3}\right)$$
$$z_1 = \sqrt[4]{t_1} = \sqrt[4]{1}\operatorname{cis}\left(\frac{\frac{-2\pi}{3}+k2\pi}{4}\right) \text{ met } k \in \{0,1,2,3\}$$
$$(z_1)_1 = \sqrt[4]{1}\operatorname{cis}\left(\frac{\frac{-2\pi}{3}}{4}\right) = \operatorname{cis}\left(\frac{-\pi}{6}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$
$$\Rightarrow (z_2)_1 = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$
$$(z_1)_2 = \sqrt[4]{1}\operatorname{cis}\left(\frac{\frac{-2\pi}{3}+2\pi}{4}\right) = \operatorname{cis}\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$
$$\Rightarrow (z_2)_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$
$$(z_1)_3 = \sqrt[4]{1}\operatorname{cis}\left(\frac{\frac{-2\pi}{3}+4\pi}{4}\right) = \operatorname{cis}\left(\frac{5\pi}{6}\right) = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$$
$$\Rightarrow (z_2)_3 = \frac{-\sqrt{3}}{2} - \frac{1}{2}i$$
$$(z_1)_4 = \sqrt[4]{1}\operatorname{cis}\left(\frac{\frac{-2\pi}{3}+6\pi}{4}\right) = \operatorname{cis}\left(\frac{4\pi}{3}\right) = \frac{-1}{2} - \frac{\sqrt{3}}{2}i$$
$$\Rightarrow (z_2)_4 = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

Er bestaan geen reële nulpunten. Als we elk nulpunt uit z_1 vermenigvuldigen

met zijn complementaire z_2 bekomen we de ontbinding in factoren in $\mathbb{R}[z]$.

$$F(x) = \left[z - \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)\right] \cdot \left[z - \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)\right] \cdot \left[z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\right] \cdot \left[z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\right]$$
$$\cdot \left[z - \left(\frac{-\sqrt{3}}{2} + \frac{1}{2}i\right)\right] \cdot \left[z - \left(\frac{-\sqrt{3}}{2} - \frac{1}{2}i\right)\right] \cdot \left[z - \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}i\right)\right] \cdot \left[z - \left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i\right)\right]$$
$$= (z^2 - \sqrt{3}z + 1) \cdot (z^2 - z + 1) \cdot (z^2 + \sqrt{3}z + 1) \cdot (z^2 + z + 1)$$

Andere methode:

$F(x) = z^8 + z^4 + 1$

$= z^8 + 2z^4 - z^4 + 1$

$= (z^4 + 1)^2 - z^4$

$= [(z^4 + 1) - z^2] \cdot [(z^4 + 1) + z^2]$

$= [z^4 + 1 + 2z^2 - 3z^2] \cdot [z^4 + 1 + 2z^2 - z^2]$

$= [(z^2 + 1)^2 - 3z^2] \cdot [(z^2 + 1)^2 - z^2]$

$= [(z^2 + 1) - \sqrt{3}z] \cdot [(z^2 + 1) + \sqrt{3}z] \cdot [(z^2 + 1) - z] \cdot [(z^2 + 1) + z]$

$= (z^2 - \sqrt{3}z + 1) \cdot (z^2 + \sqrt{3}z + 1) \cdot (z^2 - z + 1) \cdot (z^2 + z + 1)$

Ook hieruit kun je afleiden dat er geen reële nulpunten zijn. Je kunt wel 8

nulpunten vinden in $\mathbb{C}$. Dit doen we met de discrimant van de vier

resulterende 2^{de} graads functies.

$(z^2 - \sqrt{3}z + 1)$

$\Delta_1 = (-\sqrt{3})^2 - 4 \cdot 1 \cdot 1 = -1$

$$\Rightarrow (z_1)_1 = \frac{-(-\sqrt{3}) - \sqrt{-1}}{2 \cdot 1} = \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

$$\Rightarrow (z_1)_2 = \frac{-(-\sqrt{3}) + \sqrt{-1}}{2 \cdot 1} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$(z^2 + \sqrt{3}z + 1)$

$\Delta_2 = (\sqrt{3})^2 - 4 \cdot 1 \cdot 1 = -1$

$$\Rightarrow (z_2)_1 = \frac{-\sqrt{3} - \sqrt{-1}}{2 \cdot 1} = \frac{-\sqrt{3}}{2} - \frac{1}{2}i$$

$$\Rightarrow (z_2)_2 = \frac{-\sqrt{3} + \sqrt{-1}}{2 \cdot 1} = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$$

$(z^2 - z + 1)$

$\Delta_3 = (-1)^2 - 4 \cdot 1 \cdot 1 = -3$

$$\Rightarrow (z_3)_1 = \frac{-(-1) - \sqrt{-3}}{2 \cdot 1} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_3)_2 = \frac{-(-1) + \sqrt{-3}}{2 \cdot 1} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$(z^2 + z + 1)$

$\Delta_4 = 1^2 - 4 \cdot 1 \cdot 1 = -3$

$$\Rightarrow (z_4)_1 = \frac{-1 - \sqrt{-3}}{2 \cdot 1} = \frac{-1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow (z_4)_2 = \frac{-1 + \sqrt{-3}}{2 \cdot 1} = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$$

References